In [1]:
```
# We will be using stepwise subset to model our EDA
# First we will check our trimmed data for any null values
import pandas as pd
df = pd.read_csv("retail_inventory_2023_2024Trimmed.csv")
```

In [2]:
```
df.isna().sum()
```

Out[2]:
```
Date                    0
Store ID                0
Product ID              0
Category                0
Region                  0
Inventory Level         0
Units Sold              0
Units Ordered           0
Demand Forecast         0
Price                   0
Discount                0
Weather Condition       0
Holiday/Promotion       0
Competitor Pricing      0
Seasonality             0
Unnamed: 15         36600
Unnamed: 16         36600
dtype: int64
```

In [ ]:
```
# We have two columns that have no values in it and that are also unamed.
# Something in our preprocessing step messed up that format.
# We'll go ahead and drop those two columns now.
```

In [5]:
```
df = df.iloc[:, :-2]
```

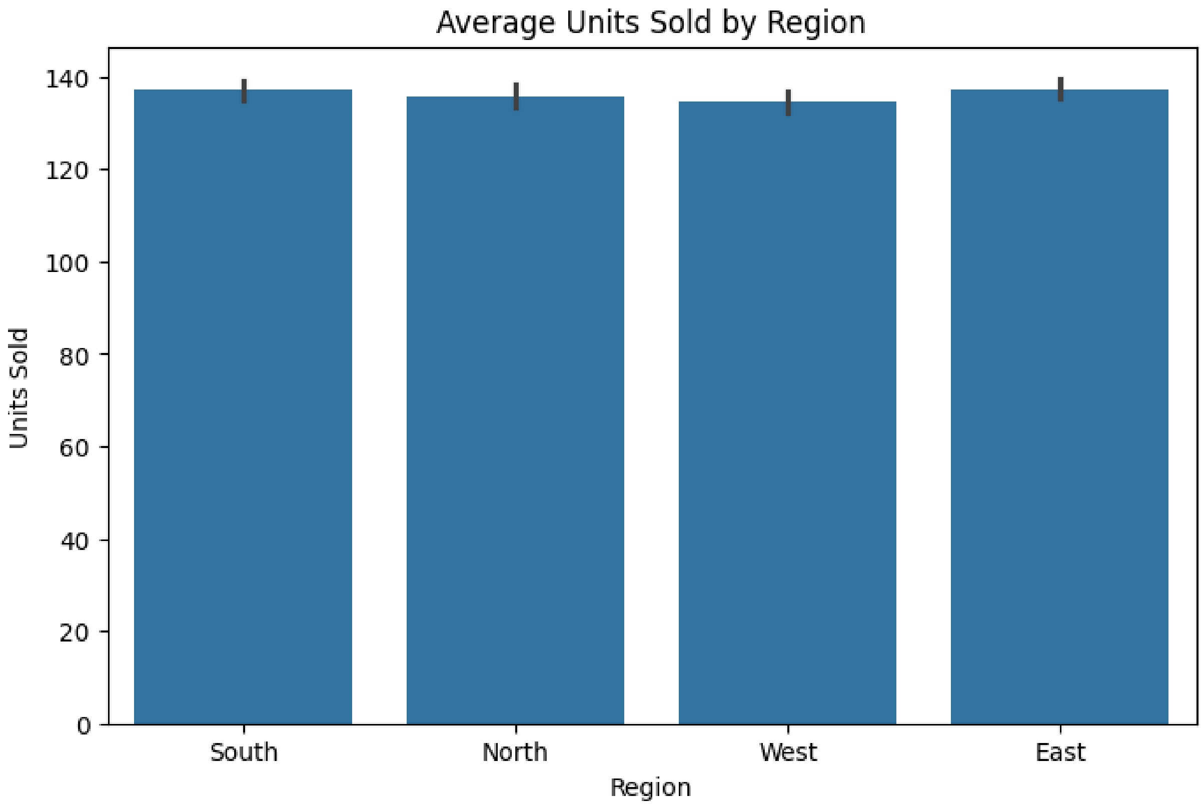
In [6]:
```
df.isna().sum()
```

Out[6]:
```
Date                  0
Store ID              0
Product ID            0
Category              0
Region                0
Inventory Level       0
Units Sold            0
Units Ordered         0
Demand Forecast       0
Price                 0
Discount              0
Weather Condition     0
Holiday/Promotion     0
Competitor Pricing    0
Seasonality           0
dtype: int64
```

In [ ]:
```
# Our dependent variable is Units Sold. We'll focus on that and observe
# EDA through bar graphs, boxplots, and scatterplots
```

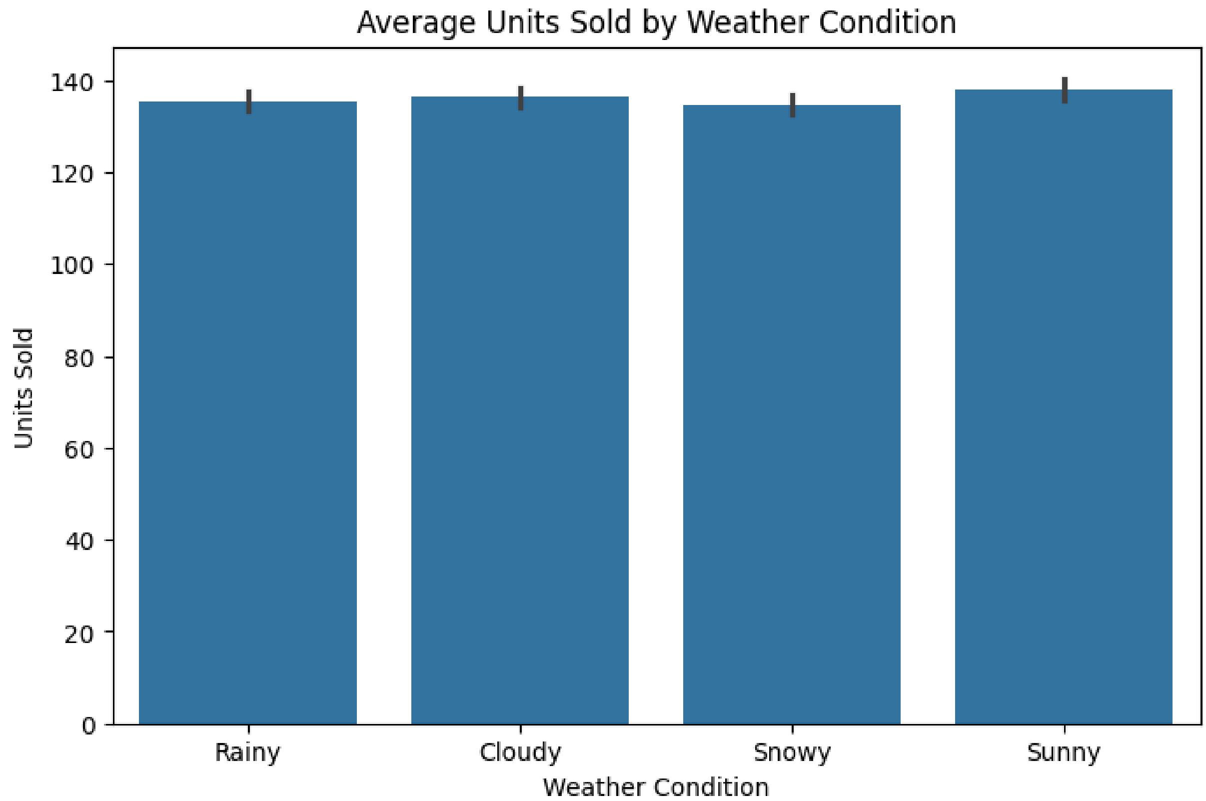
In [ ]:
```
# We are going to see if the region, weather condition, and seasonality
# have a part to play with units sold.
```

In [36]:
```
# Region
plt.figure(figsize=(8,5))
sns.barplot(x='Region', y='Units Sold', data=df, estimator='mean')
plt.title('Average Units Sold by Region')
plt.show()
```

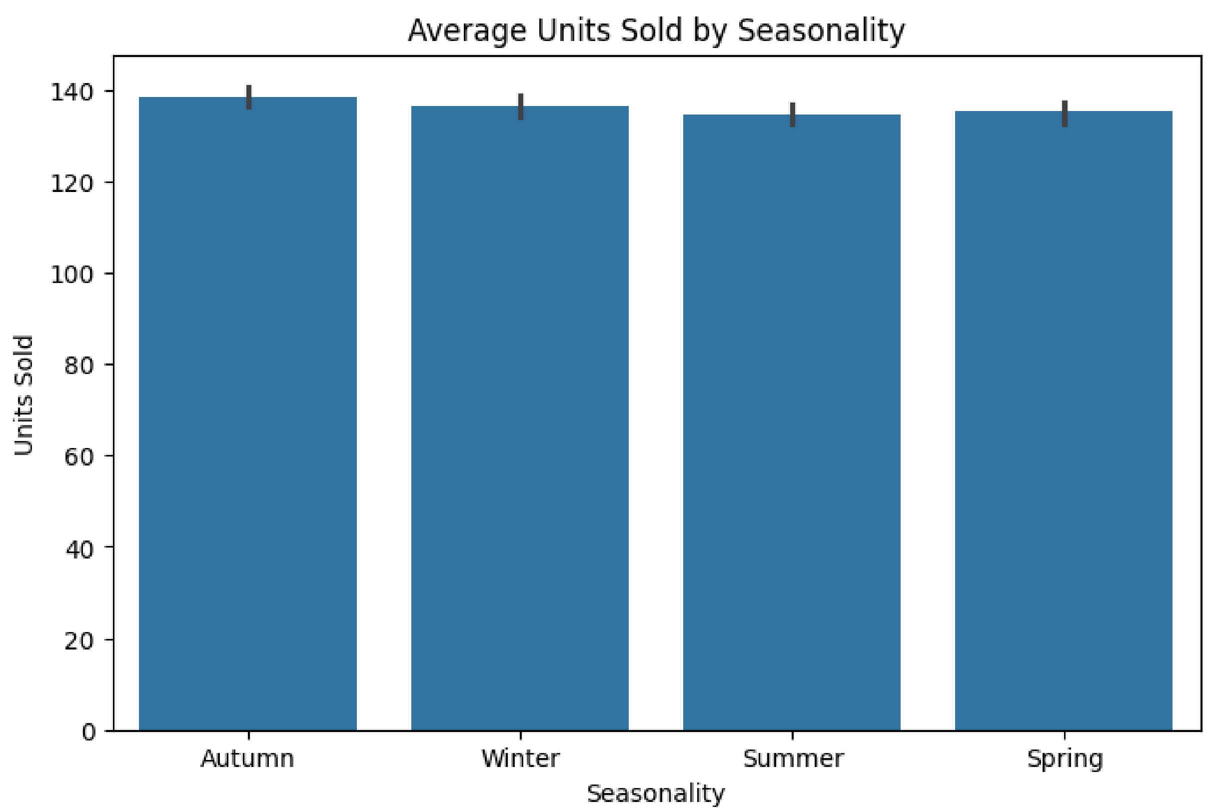


In [ ]:
```
# Weather Condition
```

In [37]:
```
plt.figure(figsize=(8,5))
sns.barplot(x='Weather Condition', y='Units Sold', data=df, estimator='mean')
plt.title('Average Units Sold by Weather Condition')
plt.show()
```



In [38]:
```
# Seasonality
plt.figure(figsize=(8,5))
sns.barplot(x='Seasonality', y='Units Sold', data=df, estimator='mean')
plt.title('Average Units Sold by Seasonality')
plt.show()
```



In [ ]:
```
# The region, weather, and season don't have an effect on units sold.
# Regardless of these conditions, customers will still
# go into the store and purchase items.
```

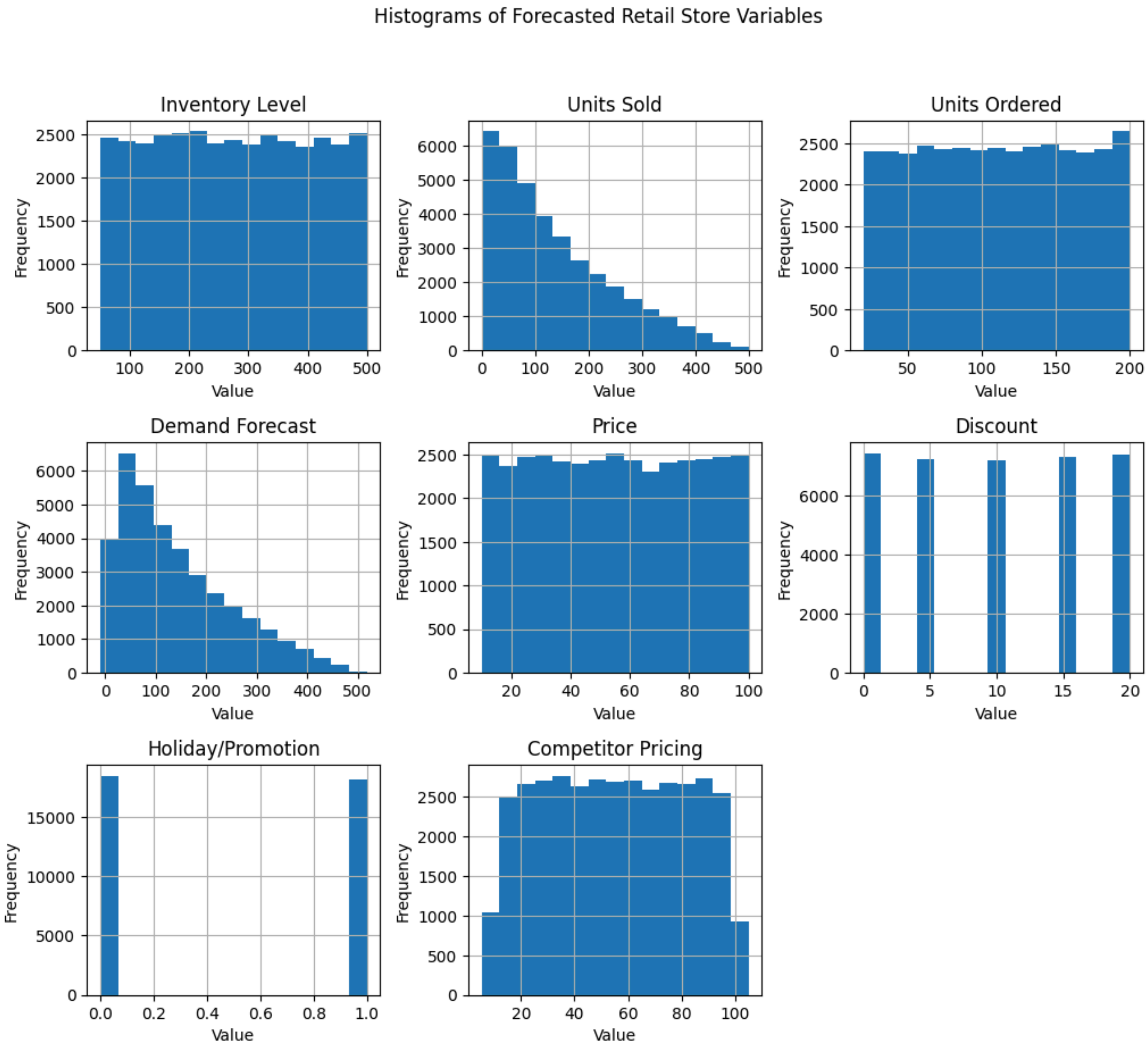
In [7]:
```
import matplotlib.pyplot as plt
import seaborn as sns

print(df.head())
print(df.info())
```

```
         Date Store ID Product ID  Category Region  Inventory Level  \
0  2023-01-01     S001      P0001  Clothing  South              425
1  2023-01-01     S001      P0002      Toys  North              240
2  2023-01-01     S001      P0003  Clothing  North              330
3  2023-01-01     S001      P0004  Clothing  North              383
4  2023-01-01     S001      P0005      Toys  North              310

   Units Sold  Units Ordered  Demand Forecast  Price  Discount  \
0         182             72           194.79  48.37        10
1          91            135            99.06  69.97        20
2         297             58           288.23  38.50        10
3          10            166            26.69  31.03        10
4          70            177            78.32  15.11         0

  Weather Condition  Holiday/Promotion  Competitor Pricing Seasonality
0             Rainy                  1               49.99      Autumn
1            Cloudy                  0               66.91      Winter
2             Snowy                  1               39.10      Summer
3             Sunny                  1               32.01      Winter
4             Rainy                  0               14.21      Winter
<class 'pandas.DataFrame'>
RangeIndex: 36600 entries, 0 to 36599
Data columns (total 15 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Date                36600 non-null  str
 1   Store ID            36600 non-null  str
 2   Product ID          36600 non-null  str
 3   Category            36600 non-null  str
 4   Region              36600 non-null  str
 5   Inventory Level     36600 non-null  int64
 6   Units Sold          36600 non-null  int64
 7   Units Ordered       36600 non-null  int64
 8   Demand Forecast     36600 non-null  float64
 9   Price               36600 non-null  float64
 10  Discount            36600 non-null  int64
 11  Weather Condition   36600 non-null  str
 12  Holiday/Promotion   36600 non-null  int64
 13  Competitor Pricing  36600 non-null  float64
 14  Seasonality         36600 non-null  str
dtypes: float64(3), int64(5), str(7)
memory usage: 4.2 MB
None
```

In [ ]:
```
# Here we can see we have a couple of numerical and categorical values in this
# data set Some categories are related to each other, like weather condition
# and Seasonality. Going to start showing some visuals to help us determine
# what might be important to keep for our multiple linear regression

# There are also 36,600 recorded units sold in the year of 2023-2024. There
# were five stores in total with recorded data. Each store that recorded data
# for a certain day had different departments ranging from toys to clothing
# to furniture. Typical departments you'd find in a store
```

In [ ]:
```
# Now time to display some histograms to understand the distribution
# and skewness
```

In [18]:
```
import matplotlib.pyplot as plt

numeric_cols = df.select_dtypes(include=['float64', 'int64']).columns
axes = df[numeric_cols].hist(figsize=(12,10), bins=15)
plt.subplots_adjust(hspace=0.4, wspace=0.3)

for ax in axes.flatten():
    ax.set_xlabel("Value")
    ax.set_ylabel("Frequency")
plt.suptitle("Histograms of Forecasted Retail Store Variables")
plt.show()
```



In [9]:
```python
print(df[numeric_cols].skew())
```

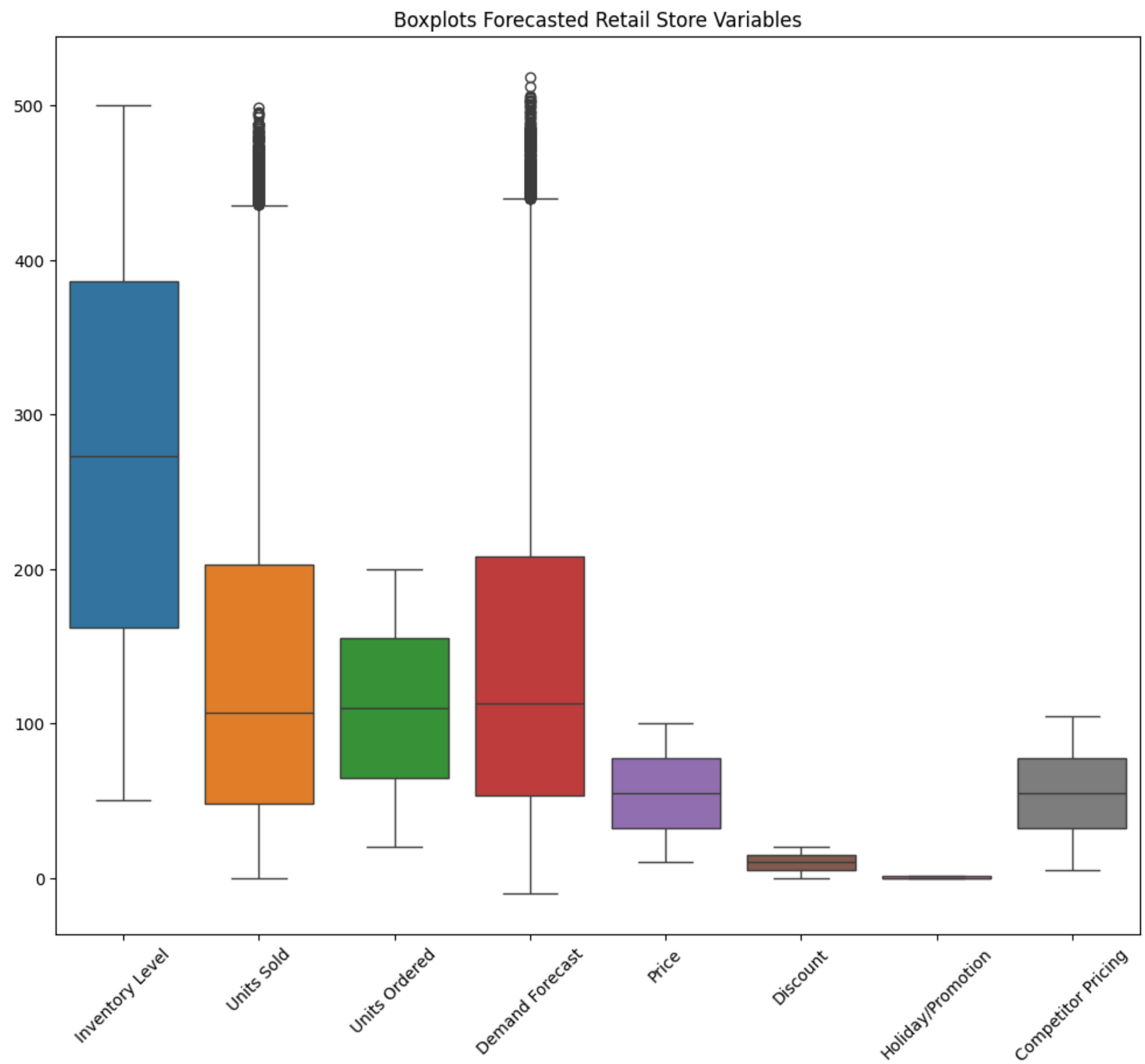
```
Inventory Level       0.011593
Units Sold            0.900578
Units Ordered        -0.000461
Demand Forecast       0.891590
Price                 0.002817
Discount             -0.002700
Holiday/Promotion     0.017378
Competitor Pricing    0.001928
dtype: float64
```

In [ ]:
```python
# The frequecy for units sold, our dependent variable, has a decreased trend
# going down. Overall this indicates the supply and demand theory of higher
# supply, less demand. The demand forecast backs up  this theory for the
# most part too with the overall decrease trend as demand reaches its
# threshold. Besides these two variables, we can see that the discount
# seems to be only in groups of 5% discounts. 5%, 10%, 15%, and 20%.
# The holiday/promotion seems to be hot encoded. It's either a
# holiday/promotion or not. Only units ordered and discount are left
# skewed based on their negative skewness. The rest are right skewed.
# Visually speaking, it seems competitor pricing is the closest to
```

```
# being a nice bell shaped curve. Making it the closest to normal
# distribution
```

In [ ]:
```
# Boxplots to detect outliers and confirm skewness
```

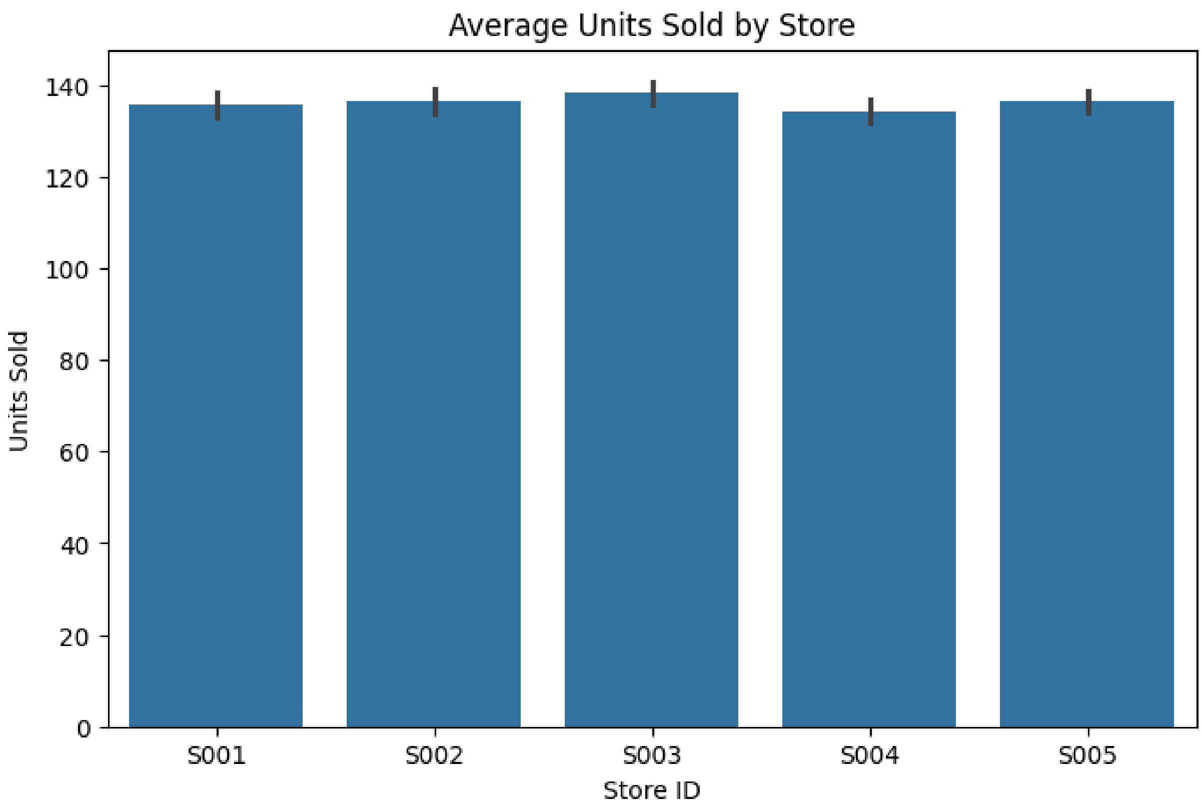
In [21]:
```
plt.figure(figsize=(12,10))
sns.boxplot(data=df[numeric_cols])
plt.title('Boxplots Forecasted Retail Store Variables')
plt.xticks(rotation=45)
plt.show()
```



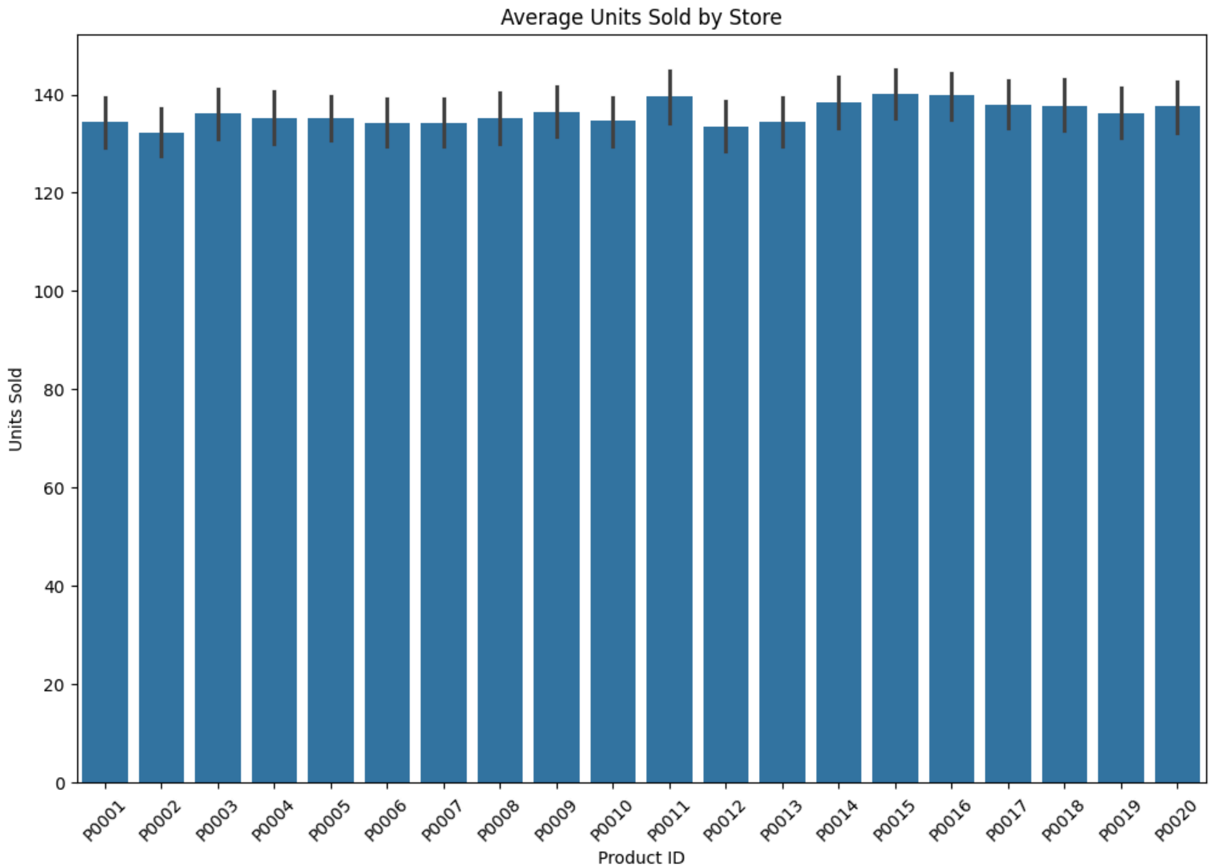
In [ ]:
```
# As we can see visually, units sold and demand forecast tend to have outliers
# while the rest don't. This is due to the nature of supply and demand. What
# can attest for these reasons is the goal of our MLR predictions. Demand
# spikes, out of stock inventory, and expired/long shelf life products cause
# these volatile issues. We wish to prevent stores from losing revenue or
# having too much inventory to prevent long shelf life, so the need for a
# good prediction is evident. It's also quite interesting to see how the
# majority of variables right skewed visually. Since they are
# gravitating towards the bottom of the quartiles.
```

In [ ]:
```
# Now it's time to view the bar graphs via different stores. We want to
# analyze the mean of units sold in each store to see how the stores
# differ. We will also take into account how each product is sold as
# well
```

In [22]:
```
plt.figure(figsize=(8,5))
sns.barplot(x='Store ID', y='Units Sold', data=df, estimator='mean')
plt.title('Average Units Sold by Store')
plt.show()
```

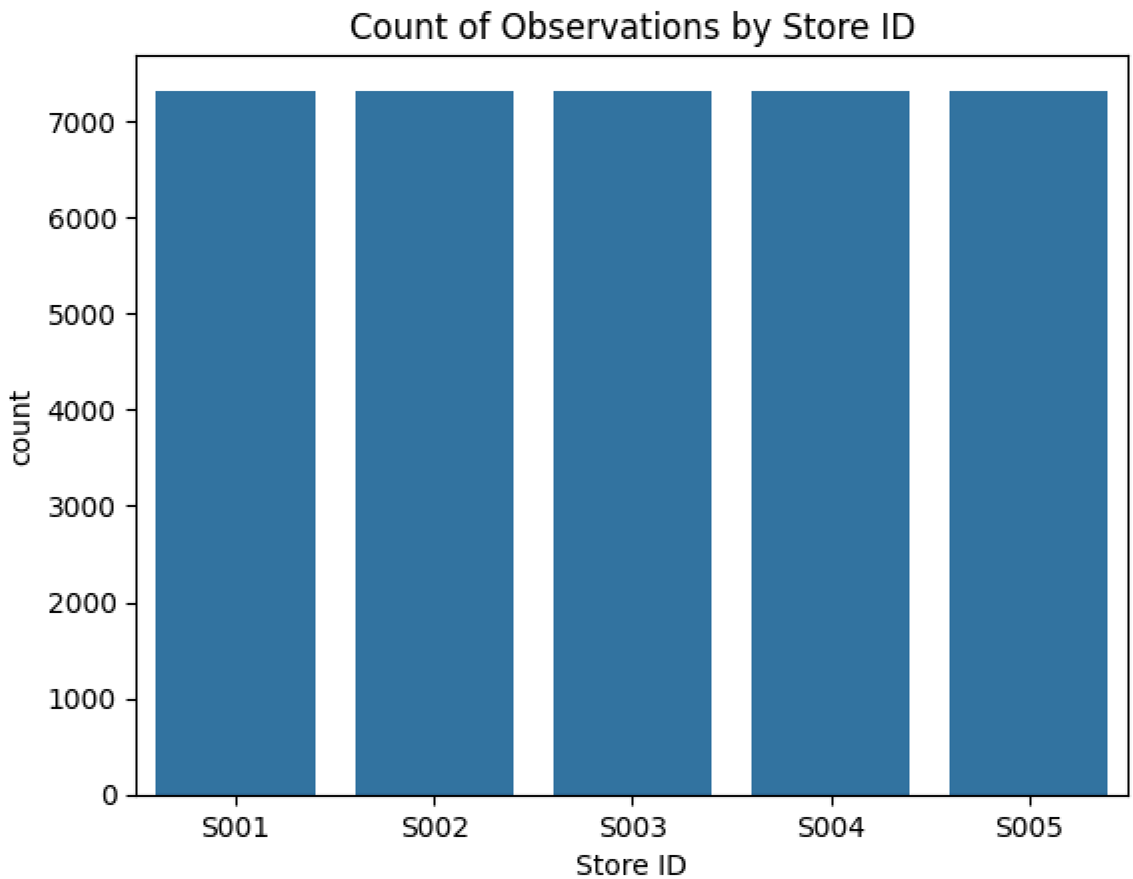


In [24]:
```
plt.figure(figsize=(12,8))
sns.barplot(x='Product ID', y='Units Sold', data=df, estimator='mean')
plt.title('Average Units Sold by Store')
plt.xticks(rotation=45)
plt.show()
```



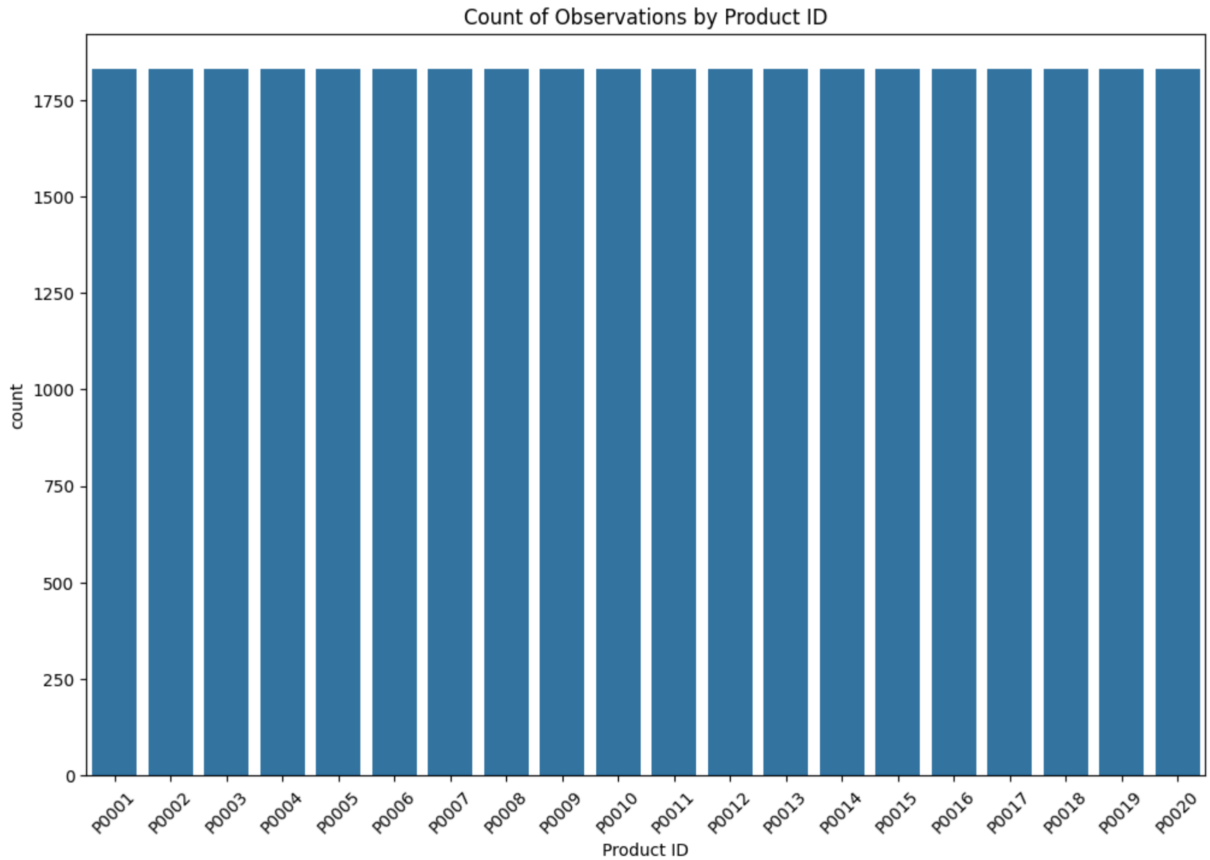
In [25]:
```
# The average units sold at each store seem relatively equal. Along with the
# products too! Our data has a roughly, good represenation of each store and
# product. To confirm, we will look at the store counts and products too.
# Just to really show our data is well represented which means our model will
# handle/predict new data accurately as well
```

In [26]:
```
sns.countplot(x='Store ID', data=df)
plt.title('Count of Observations by Store ID')
plt.show()
```



In [28]:

```
plt.figure(figsize=(12,8))
sns.countplot(x='Product ID', data=df)
plt.title('Count of Observations by Product ID')
plt.xticks(rotation=45)
plt.show()
```



In [ ]:

```
# Finally for visuals we will be making a correlation heat map. This will
# help us identify what factors have strong correlation to help us decided
# what variables we should choose for our multiple linear regression model.
# There doesn't seem to be a strong bias from either a unique store or product
```

In [30]:

```
plt.figure(figsize=(12,8))
sns.heatmap(df[numeric_cols].corr(), annot=True, cmap='RdYlGn')
plt.title('Correlation Matrix of Forecasted Retail Store Variables')
plt.show()
```

In [31]:
```
# Our focus will be units sold. here there are a lot of red or "weak" correlations
# between the majority of these variables. While looking at units sold though, we
# can see the green/light green or "stronger" relationships with inventory level
# and demand forecast. With a slightly weaker relationship between units ordered
# and holiday/promotion. This is indicating that our main focuses are going to be
# supply vs demand. With the help of external factors like holiday/promotion,
# discount, and units ordered, our multiple linear regression model is going to
# take a look at crafting the best prediction outcome based on demand of the
# product. This will be able to help us predict how much stock
# to buy because we will be able to predict how many units are going to be sold.
```

In [ ]:
```
# Finally we will run a describe analysis to view the overall numerical values
```

In [33]:
```
print(df.describe())
```

```
       Inventory Level    Units Sold  Units Ordered  Demand Forecast  \
count     36600.000000  36600.000000   36600.000000     36600.000000
mean        274.149945    136.173716     110.142678       141.208696
std         129.958751    108.784554      52.194365       109.081550
min          50.000000      0.000000      20.000000        -9.970000
25%         162.000000     48.000000      65.000000        53.487500
50%         273.000000    107.000000     110.000000       112.980000
75%         386.000000    203.000000     155.000000       207.977500
max         500.000000    499.000000     200.000000       518.550000

              Price      Discount  Holiday/Promotion  Competitor Pricing
count  36600.000000  36600.000000       36600.000000        36600.000000
mean      54.988310     10.003279           0.495656           55.012043
std       26.076935      7.102970           0.499988           26.259164
min       10.000000      0.000000           0.000000            5.340000
25%       32.397500      5.000000           0.000000           32.437500
50%       54.870000     10.000000           0.000000           54.920000
75%       77.742500     15.000000           1.000000           77.730000
max       99.990000     20.000000           1.000000          104.940000
```

In [34]:
```
# These numbers prove what we saw visually in terms of outliers and the
# distribution of dataBased on the mean and medians, we can see that the
# units ordered and discount are slightly left skewed. While the other
# variables are slightly right skewed. Units sold and demand forecast are
# the most skewed out of all the variables. Indicating most of our data is
# somewhat normally distributed. The standard deviation for our numerical
# values are higher for demand forecast and units sold versus the other
# variables. This explains the heavy skew that associates
# with demand forecast and units sold
```

In [35]:
```
# Final thoughts
```

In [ ]:
```
# Our data is well represented across all five stores and the total products that
# are being sold. Taking into consideration our high skewness with demand forecast,
# we might need to log transform our data. We are going to recommend not to use
# demand forecast. It is too highly correlated with units sold and could cause
# multicollinearity. On top of that, we would consider including inventory level,
# as well as holiday/promotion, discount, and units ordered. Given their slightly
# better correlation compared to the other variables.
# Finally the categorical variables, region, weather conditions, and seasonality,
# don't seem relevant. Demand of these products at all five stores still holds
# true regardless of region, weather, or season. There is a slight favorability
# towards the east region, sunny day, and autumn weather, but it's not enough
# to have an impact on units sold.
```